

*Non-Calculator*

On my honor, before God, I have neither given, nor received any aid on this quiz.

Signature \_\_\_\_\_

**You have 30 minutes.**

**SHOW ALL WORK FOR FULL CREDIT!**

**1.** Identify each number as *real*, *complex*, *pure imaginary*, or *nonreal complex*. **(2 pts)**  
(More than one of these descriptions may apply.)

a.  $-\sqrt{25}$

b.  $2i\sqrt{3}$

**2.** Find the quotient. Write your answer in standard form. **(2 pts)**

$\sqrt{-54} / \sqrt{6}$

**3.** Simplify each power of  $i$ . **(2 pts)**

a.  $i^{30}$

b.  $1 / i^9$

**4.** Is  $3 - 2i$  a solution of the equation? Show work.

YES / NO

**(2 pts)**

$$x^2 - 6x + 13 = 0$$

**5.** Find the product. Write your answer in standard form.

**(2 pts)**

$$(2 - 7i)^2$$

**6.** Find the quotient. Write your answer in standard form.

**(3 pts)**

$$(5 - i) / (3 + 2i)$$

7. Solve the equation over the set of complex numbers.

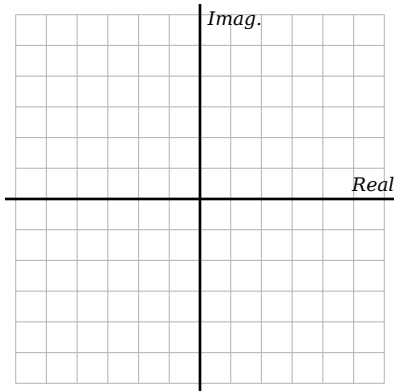
(4 pts)

$$3x^2 + 5 = 2x$$

8. Find the sum of  $-4 + i$  and  $1 - 5i$ . Write your answer in standard form.

(3 pts)

Graph both complex numbers and their resultant. Then find the absolute value of the sum.



9. Write the complex number in rectangular form.

(3 pts)

$$4(\cos 5\pi/6 + i \sin 5\pi/6)$$

**10.** Write each complex number in trigonometric (polar) form, with  $\theta$  in the interval  $[0^\circ, 360^\circ)$ .

**(4 pts)**

a.  $1 - i\sqrt{3}$

b.  $-7i$

**11.** The Julia set: compute  $z^2 - 1$ ,  $(z^2 - 1)^2 - 1$ ,  $[(z^2 - 1)^2 - 1]^2 - 1$ , ... If the absolute value of any result exceeds 2,  $z$  is **not** in the Julia set. Otherwise it is.

**(3 pts)**

Is  $z = \frac{1}{2}i$  in the Julia set? Show work.      YES / NO

## Answer Key • Practice Quiz B

Every step shown. Box = final answer.

1. (2 pts)

- 1) a.  $-\sqrt{25} = -5$ . So  $b = 0$ : it is real (and every real number is complex).
- 2) b.  $2i\sqrt{3} = 0 + 2\sqrt{3}i$ .  $a = 0$  and  $b \neq 0$ : pure imaginary, which is also nonreal complex.

**a. complex, real    b. complex, pure imaginary, nonreal complex**

2. (2 pts)

- 1) Pull out  $i$  FIRST:  $\sqrt{-54} = i\sqrt{54} = i\sqrt{9 \cdot 6} = 3i\sqrt{6}$ .
- 2)  $3i\sqrt{6} / \sqrt{6} = 3i$ .

**3i**

3. (2 pts)

- 1) a.  $30 \div 4 = 7$  remainder 2, so  $i^{30} = i^2 = -1$ .
- 2) b.  $9 \div 4 = 2$  remainder 1, so  $i^9 = i$ .
- 3)  $1/i \cdot (-i)/(-i) = -i / (-i^2) = -i / 1 = -i$ .

**a. -1    b. -i**

4. (2 pts)

- 1)  $(3 - 2i)^2 = 9 - 12i + 4i^2 = 9 - 12i - 4 = 5 - 12i$ .
- 2)  $-6(3 - 2i) = -18 + 12i$ .
- 3) Add everything:  $(5 - 12i) + (-18 + 12i) + 13 = 0$ . It equals 0.

**YES**

5. (2 pts)

- 1)  $(2 - 7i)(2 - 7i) = 4 - 14i - 14i + 49i^2$ .
- 2)  $49i^2 = -49$ .
- 3)  $4 - 49 - 28i$ . (Squaring each part separately,  $4 + 49i^2$ , loses the  $-28i$ .)

**-45 - 28i**

6. (3 pts)

- 1) Multiply top and bottom by the conjugate of the bottom,  $3 - 2i$ .
- 2) Top:  $(5 - i)(3 - 2i) = 15 - 10i - 3i + 2i^2 = 15 - 13i - 2 = 13 - 13i$ .
- 3) Bottom:  $(3 + 2i)(3 - 2i) = 9 + 4 = 13$ .
- 4)  $(13 - 13i)/13 = 1 - i$ .

**1 - i**

7. (4 pts)

- 1) Set equal to 0:  $3x^2 - 2x + 5 = 0$ , so  $a = 3$ ,  $b = -2$ ,  $c = 5$ .
- 2) Discriminant:  $(-2)^2 - 4(3)(5) = 4 - 60 = -56$ .
- 3)  $\sqrt{-56} = i\sqrt{56} = 2i\sqrt{14}$ , so  $x = (2 \pm 2i\sqrt{14}) / 6$ .
- 4) Split and reduce:  $2/6 = 1/3$  and  $2\sqrt{14}/6 = \sqrt{14}/3$ .

**$x = 1/3 \pm (\sqrt{14}/3)i$**

**8.** (3 pts)

- 1) Add:  $(-4 + 1) + (1 - 5)i = -3 - 4i$ .
- 2) Graph  $-4 + i$  to  $(-4, 1)$ ,  $1 - 5i$  to  $(1, -5)$ , and the resultant (the parallelogram diagonal) to  $(-3, -4)$ .
- 3)  $|-3 - 4i| = \sqrt{(-3)^2 + (-4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5$ .

$$\mathbf{-3 - 4i, \quad |-3 - 4i| = 5}$$

**9.** (3 pts)

- 1) Convert:  $5\pi/6 \cdot 180^\circ/\pi = 150^\circ$ , quadrant II, reference angle  $30^\circ$ .
- 2)  $\cos 150^\circ = -\sqrt{3}/2$ ,  $\sin 150^\circ = 1/2$ .
- 3)  $4(-\sqrt{3}/2) + 4(1/2)i = -2\sqrt{3} + 2i$ .

$$\mathbf{-2\sqrt{3} + 2i}$$

**10.** (4 pts)

- 1) a. Sketch:  $x = 1$ ,  $y = -\sqrt{3}$  is quadrant IV.
- 2)  $r = \sqrt{1 + 3} = \sqrt{4} = 2$ .
- 3)  $\tan \theta = -\sqrt{3}/1$ , reference angle  $60^\circ$ . Quadrant IV:  $\theta = 360^\circ - 60^\circ = 300^\circ$ .
- 4) b.  $-7i = 0 - 7i$  lies on the negative imaginary axis:  $r = 7$ ,  $\theta = 270^\circ$ .

$$\mathbf{a. 2(\cos 300^\circ + i \sin 300^\circ) \quad b. 7(\cos 270^\circ + i \sin 270^\circ)}$$

**11.** (3 pts)

- 1)  $z^2 = (\frac{1}{2}i)^2 = \frac{1}{4}i^2 = -\frac{1}{4}$ .  $z^2 - 1 = -5/4$ .  $|-5/4| = 5/4$ , not over 2.
- 2)  $(-5/4)^2 - 1 = 25/16 - 16/16 = 9/16$ .  $|9/16| = 9/16$ , not over 2.
- 3)  $(9/16)^2 - 1 = 81/256 - 256/256 = -175/256$ .  $|-175/256|$  is less than 1.
- 4) No result exceeds 2 (the values stay between  $-5/4$  and 1).

**YES**